Synthack Presents

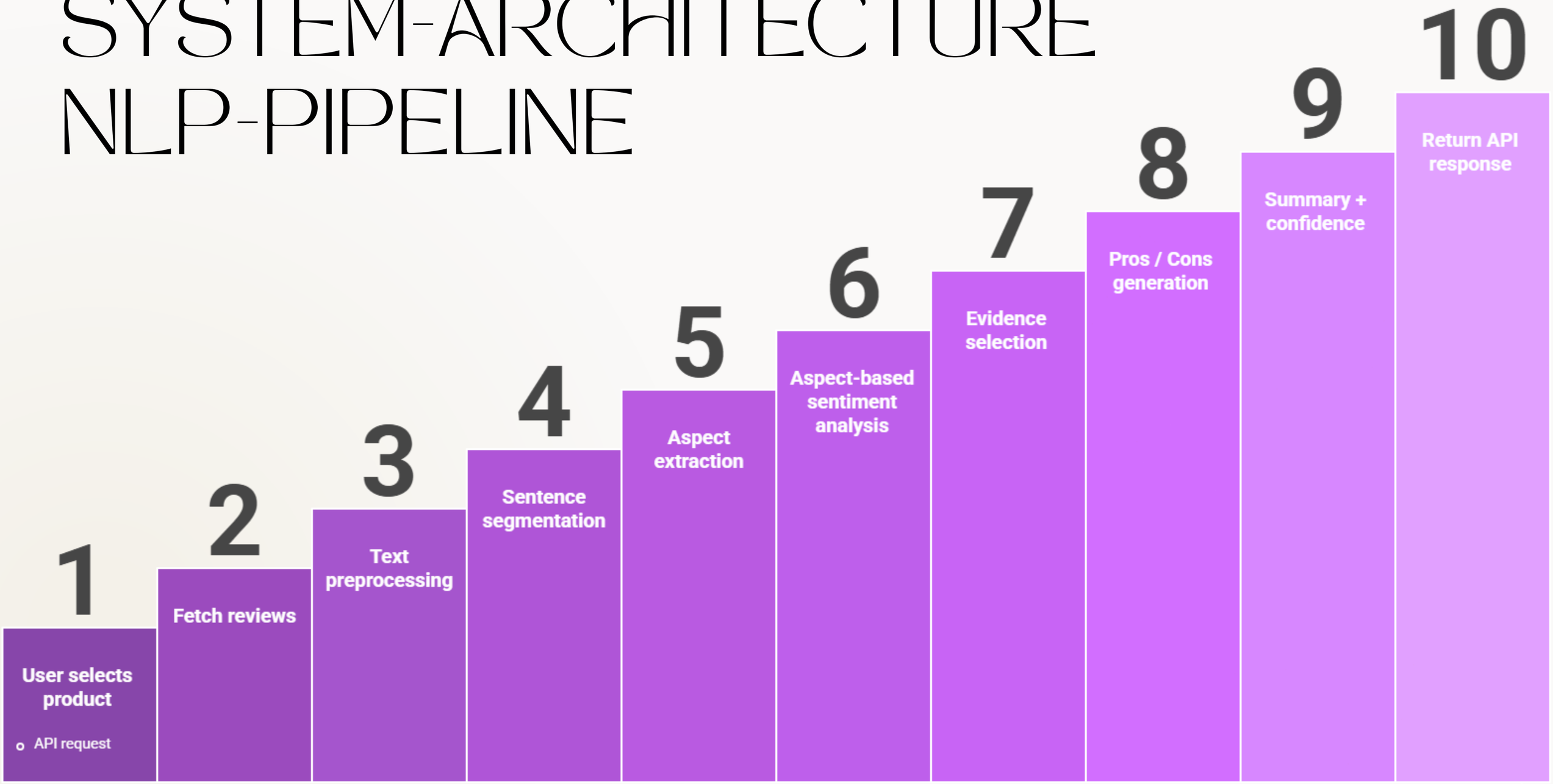
REVIEWLYTICS

PROBLEM STATEMENT: TASK-5

SOLUTION

- Automatically analyzes product reviews
- We use nlp to identify the sentiment of each review and then map it with the product
- Extracts top aspects customers discuss
- Performs aspect-based sentiment analysis
- Generates pros and cons with evidence
- Provides confidence scores and summaries
- Reviews → NLP Engine → Insights

SYSTEM-ARCHITECTURE NLP-PIPELINE



Made with Napkin

Models Used :

Model	Purpose
DistilBERT	used for sentiment classification: ie., defines whether a sentence is positive or negative
SentenceTransformer	it is used to make vector embeddings of sentences
DeBERTa ABSA	is used for determining sentiment specifically toward particular aspect
BERTopic	is used to discover new topics directly from the reviews

Output based on different level of nlp models

KEYWORD METHOD

Durability

Positively mentioned in 1 review(s).

> *"Between me, my kids, and my husband, this lasted only two weeks"*

TRANSFORMERS METHOD

Durability

Negatively mentioned in 1 review(s).

> *"Between me, my kids, and my husband, this lasted only two weeks"*

Without ABSA

CONS

Battery Life

Negatively mentioned in 1 review(s).

> *"camera quality is really nice but battery life is short"*

Build Quality

Negatively mentioned in 2 review(s).

> *"display quality is not good"*

Camera

Negatively mentioned in 1 review(s).

> *"camera quality is really nice but battery life is short"*

With ABSA

PROS

Build Quality

Positively mentioned in 2 review(s).

> *"Camera quality is really nice but battery life is short"*

Camera

Positively mentioned in 1 review(s).

> *"Camera quality is really nice but battery life is short"*

Durability

Positively mentioned in 1 review(s).

> *"it lasted for long time"*

CONS

Battery Life

Negatively mentioned in 1 review(s).

> *"Camera quality is really nice but battery life is short"*

Display/Screen

Negatively mentioned in 1 review(s).

> *"display quality is not good"*

Why Noi LLM!!?

LLM Wrapper	Our System
High cost per query	Runs locally
Slow inference	Fast API
Hallucination risk	Evidence-based
Hard to Scale	Efficient pipeline

How we met the problem statement requirements

Requirement	Our Implementation
Aspect extraction	keyword + semantic similarity
Aspect sentiment	Transformer model
Pros & Cons	Evidence based selection
Summary	Aspect aggregation
Confidence	Based on review count
API Validation	Input checks

DEMO

QUESTIONS?